PEACHES and PEANUTS: affine-invariant ensemble HMC that scales, preconditions, and tolerates short warmup

PEACHES (ensemble ChEES-HMC) and PEANUTS (ensemble NUTS) are gradient-based affine-invariant ensemble samplers: a cloud of walkers runs Hamiltonian dynamics preconditioned by the *empirical covariance of the live ensemble*, so the metric is read off the walkers each iteration rather than fit during a separate warmup. This note isolates three clean, reproducible properties on controlled targets (Gaussian and Rosenbrock, exact moments known), each with seed-to-seed statistical fluctuation shown:

1. **Dimensional scaling.** On the isotropic Gaussian, per-gradient efficiency follows the optimal-HMC law ESS/grad $\sim d^{-1/4}$, *exponentially* better than the d^{-1} of the gradient-free affine sampler `emcee` (§1).
2. **Conditioning *and* curvature.** A clean 2×2: PEACHES/PEANUTS handle both, while each competitor handles only one. On the ill-conditioned Gaussian they beat HMC/NUTS run *without* mass adaptation by up to $\sim 6 \times 10^4\times$ (the ensemble preconditions for free); on the curved Rosenbrock they stay converged while `emcee` — fine on the affine Gaussian — fails to converge (§2).
3. **Warmup robustness.** PEACHES/PEANUTS are flat in warmup length, whereas NUTS needs a long warmup to fit its mass matrix and is 3–4 orders less efficient at short warmup (§3).

All efficiencies are production-only ESS per target-gradient (ArviZ rank-normalized split ESS); the gradient ensemble samplers (PEACHES/PEANUTS) use $n_{\text{walkers}} \propto d$ for a full-rank preconditioner — exactly $4d$ in the Gaussian scaling sweep, $\max(128, 4d)$ in the conditioning and warmup experiments, and 256–512 in the harder Rosenbrock scaling (see Reproducibility). Bands/error bars are the seed-to-seed spread on a fixed target.

1 Dimensional scaling: $d^{-1/4}$ vs. d^{-1}

Optimally-tuned HMC on a d-dimensional product target needs a step size $\epsilon \sim d^{-1/4}$ to hold a fixed acceptance, hence $L \sim d^{1/4}$ leapfrog steps per proposal and ESS/grad $\sim d^{-1/4}$ (Beskos et al. 2013). With acceptance pinned at 0.8 and the step size adapted, both ensemble samplers realise this law in the asymptotic tail: over $d \in [64, 256]$ the fitted efficiency slopes are -0.20 (PEACHES) and -0.21 (PEANUTS) against the theoretical -0.25, and the PEACHES trajectory length grows as $d^{+0.21}$ ($\bar{L} : 7.1 \to 9.4$ as $d : 64 \to 256$). Below $d \sim 16$ the curves are pre-asymptotic: with only $4d$ walkers the ensemble covariance is estimated from few samples, so efficiency *rises* with d to a plateau near $d \sim 32$ before the step-size law takes over — the full small-d story, invisible in a high-d-only plot.

`emcee`'s stretch move carries no gradient information; its acceptance and step degrade much faster with dimension, giving ESS/grad $\sim d^{-1}$ (fitted slope -1.26 over the converged $d \in [8, 64]$).

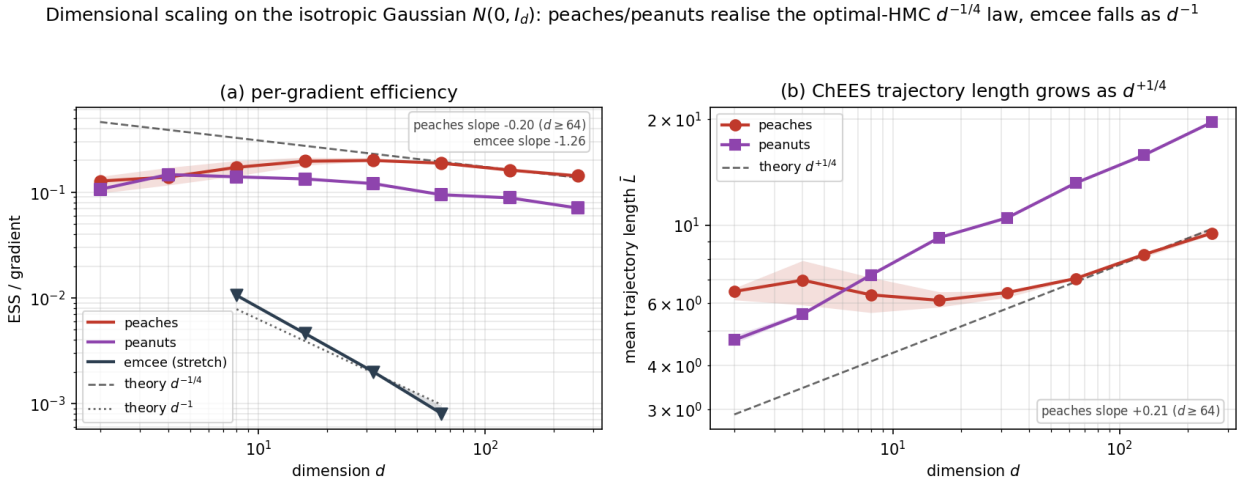


Figure 1: Dimensional scaling on the isotropic Gaussian $N(0, I_d)$, extended down to $d = 2$ and run $10\times$ longer. Uniform protocol for the gradient samplers: $n_{\text{walkers}} = 4d$ (no floor), target_accept $= 0.8$, stretch pre-warmup $w_s = 100$, warmup 4000, production $N = 3\times10^4$ (10^4 at d=256 for device memory — per-gradient efficiency is a ratio, insensitive to production length once converged). `emcee` uses $n_{\text{walkers}} = \max(128, 4d)$ and is run much longer (warmup 2×10^4, production 10^5) as it mixes far more slowly. Lines are the median over the converged seeds (6 at $d \leq 16$, 3 at $d \geq 32$), bands the seed min–max range (statistical fluctuation on a fixed target); all points converged, rank-normalized split $\hat{R} \leq 1.03$. **(a)** Per-gradient efficiency: PEACHES/PEANUTS approach the optimal-HMC law $d^{-1/4}$ (dashed; fitted slope $-0.20/-0.21$ over $d \in [64, 256]$), while `emcee` (stretch) falls as d^{-1} (dotted, slope -1.26). The gentle rise for $d \lesssim 16$ is pre-asymptotic — at $4d$ walkers the empirical ensemble covariance is noisier in few dimensions. `emcee` is shown only to d=64; beyond that it cannot be converged in a feasible run. **(b)** The mechanism: the ChEES-adapted mean trajectory length grows as $d^{+1/4}$ (PEACHES slope $+0.21$), the signature of the $\epsilon \sim d^{-1/4}$ step-size law.

The gap widens exponentially in d: at d=64 PEACHES is already $\sim$235$\times$ more efficient per gradient, and `emcee` cannot be converged at all beyond d=64 in a feasible run — itself the point.

We also examined scaling on the *paired* Rosenbrock (Fig. 2), formally a product of $d/2$ identical banana pairs and so in principle subject to the same $d^{-1/4}$ law. It is not realised over the accessible range $d \leq 32$. The per-block curvature gives the ChEES integrator no single good integration length: the adapted $\bar{L}$ is large and *bimodal* — some seeds find the long trajectory that traverses a banana ($\bar{L} \sim 200$, IAT ~ 1), others a short one ($\bar{L} \sim 20$, IAT ~ 25) — versus the Gaussian's smooth $\bar{L} \sim 6$–9. With enough walkers (256–512) and a long warmup (≥ 12000) *every* seed converges at $d \geq 8$ ($\hat{R} \leq 1.01$). But convergence is not efficiency: a converged seed that landed in the short-$\bar{L}$ mode still mixes slowly (IAT ~ 25 at the same $\hat{R}$), so the per-gradient efficiency is meaningful yet seed-variable — the band reflects the two $\bar{L}$ modes among the *converged* seeds, not unconverged runs (at d=32 all seeds find the long mode and the spread collapses). It stays low (ESS/grad $\sim 10^{-3}$) and drifts *upward* with d rather than following the law. The single/double-banana cases $d = 2, 4$ remain partly pathological even at 512 walkers (5/6 seeds converge). The robust statement on curvature is thus the qualitative one of §2: PEACHES/PEANUTS stay efficient and converge on the banana where `emcee` cannot — not any clean power law.

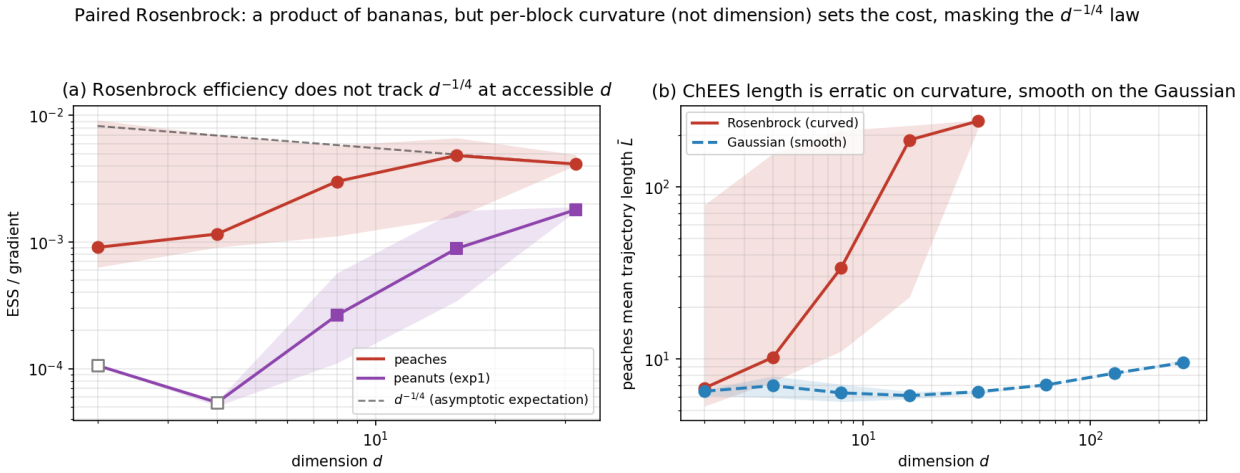


Figure 2: Why the paired Rosenbrock does not realise $d^{-1/4}$ over accessible d, despite being a product of identical banana pairs. Median over 6 seeds (PEACHES, tuned runs: 256 walkers for $d \geq 8$, 512 for $d = 2, 4$, warmup ≥ 12000) / 3 seeds (`peanuts`, `exp1`); band = seed min–max over the *converged* seeds only; hollow = a minority of seeds converged. **(a)** ESS/gradient *rises* with d (it does not track the $d^{-1/4}$ reference): at low d the extreme single-banana curvature and the short-$\bar{L}$ ChEES mode dominate. **(b)** The mechanism — PEACHES' adapted trajectory length is large and erratic on the curved banana ($\bar{L} \sim 10$–220) but smooth and slowly-growing on the Gaussian ($\bar{L} \sim 6$–9); the curvature, not the dimension, sets the cost.

2 Conditioning: the ensemble preconditions for free

The two targets probe two *different* failure modes, and the data (Fig. 3) gives a clean 2×2: PEACHES/PEANUTS handle both; each competitor handles only one.

Ill-conditioned Gaussian (the *conditioning* test). The empirical-covariance preconditioner makes PEACHES/PEANUTS exactly affine-invariant: their ESS / grad is flat in κ (≈ 0.19 / ≈ 0.10 across $\kappa = e^{2\text{–}8}$). `emcee` is likewise κ-invariant (its stretch move is affine), just dimensionally penalised ($\approx 10^{-3}$ at d=50). HMC/NUTS run with an *identity* mass matrix has no such invariance: its step size is throttled by the stiffest direction, so ESS / grad falls from 3.1×10^{-2} at $\kappa = e^2$ to 3.0×10^{-6} at $\kappa = e^8$ — *four* orders of magnitude, ending $\sim 6 \times 10^4$ below the ensemble samplers (and failing to converge, $\hat{R} = 1.7$, at $\kappa = e^8$). This conditioning is exactly the work NUTS normally does in its warmup-dependent mass adaptation; the ensemble does it for free, every iteration.

Rosenbrock (the *curvature* test). Here the challenge is not global conditioning but position-dependent curvature, which no fixed linear map can straighten. `emcee`'s single affine stretch move cannot cross the curved ridge: even at 10^5 samples it *fails to converge* for $d \geq 4$ (rank-normalized split $\hat{R} = 1.04$ at d=4, already past 1.03, then 1.14, 1.40, 2.0 at d=8, 16, 32; its apparent ESS / grad is meaningless and is greyed). PEACHES/PEANUTS integrate *local* Hamiltonian dynamics along the ridge and stay converged and efficient (ESS/grad $\approx 10^{-3}$, $\hat{R} \leq 1.03$). No-mass NUTS, notably, *does* converge here: Rosenbrock's global conditioning is mild, so the identity metric is not the bottleneck — which is exactly why this panel separates the curvature failure (`emcee`) from the conditioning failure (no-mass HMC, panel a). Only the ensemble samplers clear both.

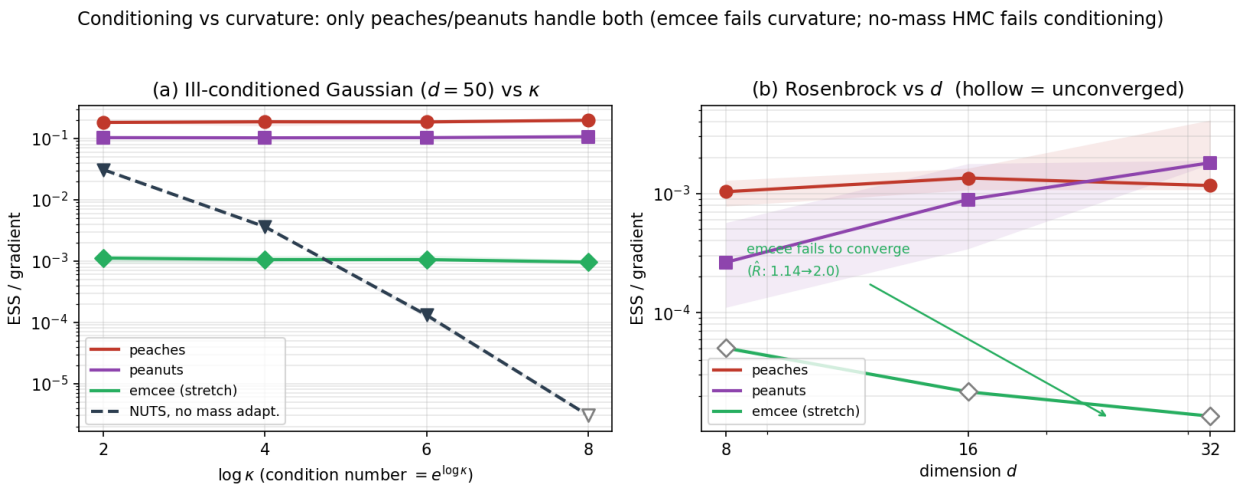


Figure 3: Robustness to conditioning and curvature. Median over 3 seeds, shaded bands the seed min–max range; **solid markers are converged** ($\hat{R} \leq 1.03$), **hollow are not** (their ESS / grad is not a meaningful efficiency). Warmup/production (Gaussian / Rosenbrock): PEACHES/PEANUTS warmup 1000 / 5000, production 5000 / 8000; `emcee` warmup $1 / 2 \times 10^4$, production $6 / 10 \times 10^4$ (run long to give it a fair chance); no-mass NUTS 1000/2000 (Gaussian). Ensemble $n_{\text{walkers}} = \max(128, 4d)$, stretch pre-warmup w_s=200 / 1000. **(a)** ill-conditioned Gaussian (d=50) vs. κ. PEACHES/PEANUTS and `emcee` are flat in κ (affine-invariant); NUTS *without* mass adaptation (identity metric) degrades sharply, collapsing and failing to converge by $\kappa = e^8$. **(b)** Rosenbrock vs. dimension (peaches/peanuts vs. `emcee`; no-mass NUTS omitted — it converges on Rosenbrock and is the panel-(a) story). PEACHES/PEANUTS stay efficient and converged; `emcee` *fails to converge* for $d \geq 4$ even at 10^5 samples (its single affine move cannot cross the curved ridge).

3 Warmup robustness

Because the preconditioner is recomputed from the live ensemble every iteration, PEACHES/PEANUTS need no mass-matrix warmup: on the ill-conditioned Gaussian their ESS / grad is essentially flat (≈ 0.16–0.21 PEACHES, ≈ 0.089 PEANUTS) across warmup $\in \{50, \ldots, 1000\}$ (Fig. 4, left). NUTS-dense, which must fit a dense mass matrix during warmup, is erratic and far below at short warmup — ESS/grad $\approx 2.4 \times 10^{-5}$ at warmup 50, a factor $\sim 10^4$ below PEACHES — and only becomes competitive once warmup ≥ 500. The same pattern holds on Rosenbrock (right): the ensemble samplers are warmup-insensitive while NUTS recovers only at long warmup. The ensemble methods need only a short gradient-free stretch pre-warmup (w_s=200) to equilibrate the walker cloud's *shape* — which costs no gradients and uses no target covariance.

4 Reproducibility

Targets: isotropic Gaussian $N(0, I_d)$ (scaling), ill-conditioned Gaussian $\kappa = e^{2\text{–}8}$ rotated (conditioning/warmup), and paired Rosenbrock (conditioning and scaling). PEANUTS uses max-tree-depth 10; the NUTS baselines are `numpyro` NUTS — "no-mass" is `adapt_mass_matrix=False` (identity metric, step size still adapted, max-tree-depth 8) and the warmup baseline is dense-mass. Run lengths and the stretch pre-warmup w_s differ by experiment (the samplers are robust to both, §3); 3–6 seeds. Per-dimension efficiency is the median over the *converged* seeds ($\hat{R} \leq 1.03$); a point is

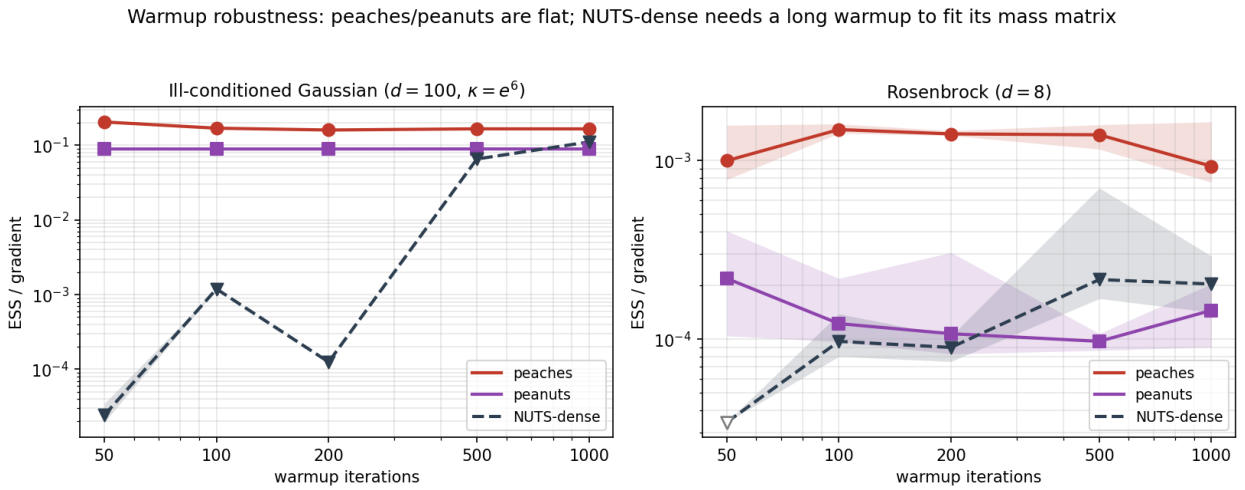


Figure 4: Warmup robustness. Median over the converged seeds (8 sampling seeds on a fixed target) with the band the interquartile range (25–75%); the Rosenbrock panel uses 256 walkers (the Gaussian 400), and we use the IQR rather than min–max because ChEES's integration length is bimodal on the banana (some seeds find the long trajectory that traverses it, others a short one — both converge, but min–max would span an order of magnitude). Production 5000 samples. ESS / grad vs. the gradient-sampler warmup length for the ill-conditioned Gaussian (d=100, $\kappa = e^6$) and Rosenbrock (d=8). PEACHES/PEANUTS are flat from warmup 50 to 1000; NUTS-dense needs $\gtrsim$500 warmup steps to fit its mass matrix and is orders of magnitude less efficient below that.

plotted solid only if a majority of its seeds converged (hollow otherwise), and stuck seeds never enter the median. All efficiencies are production-only ESS/grad.

Scaling (§1): isotropic d=2–256; gradient samplers at uniform $n_{\text{walkers}} = 4d$ (no floor), target-accept 0.8, w_s=100, warmup 4000, 3×10^4 production (10^4 at d=256 for device memory); `emcee` $n_{\text{walkers}} = \max(128, 4d)$, warmup 2×10^4, 10^5 production (tags `scaleL`/`scaleLb`, `exp4c`); Gaussian low-d ($d \le 16$) uses 6 seeds. The Rosenbrock scaling (Fig. 2) uses PEACHES at 256 walkers ($d \ge 8$) / 512 (d=2, 4), warmup 12000 / 15000, target-accept 0.8, w_s=100, 6 seeds (tag `scaleLd`); `peanuts` reuses `exp1`.

Conditioning (§2): Gaussian d=50 vs κ and Rosenbrock $d \in \{8, 16, 32\}$; ensemble $n_{\text{walkers}} = \max(128, 4d)$. PEACHES/PEANUTS warmup 1000 / 5000, production 5000 / 8000, w_s=200 / 1000 (Gaussian / Rosenbrock); `emcee` warmup 1–2×10^4, production 6–10×10^4; no-mass NUTS warmup 1000, production 2000 (tags `exp2`/`exp1`, `idmass`).

Warmup (§3): Gaussian d=100 ($\kappa = e^6$) and Rosenbrock d=8; gradient-sampler warmup $\in \{50, 100, 200, 500, 1000\}$, production 5000, w_s=200, dense-mass NUTS, 8 seeds. Gaussian $n_{\text{walkers}} = 400$ (tag `exp3`); Rosenbrock $n_{\text{walkers}} = 256$ (tag `exp3b`, bumped from 128 for cleaner short-warmup convergence). Band is the interquartile range over converged seeds.

5 Summary

On controlled targets, PEACHES and PEANUTS combine the strengths their competitors have only separately: the $d^{-1/4}$ dimensional scaling of well-tuned HMC (`emcee` scales as d^{-1}), the κ- and curvature-robustness that gradient HMC gets only *with* mass adaptation (and `emcee` gets only on affine problems), and insensitivity to warmup length (NUTS needs a long warmup). The com-

mon cause is the empirical-covariance ensemble preconditioner: it supplies the metric that HMC otherwise has to learn during warmup, for free and every iteration.